RÉSUMÉ–JOB MATCHING & TALENT MARKETPLACE ENGINE

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Department: Department of Computer Science and Information Technology | **Institution:** K L (Deemed to be) University
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Abstract

- A Java-based talent marketplace connecting candidates and recruiters.
- Candidates can register, maintain profiles, browse open jobs, view compatibility scores, identify matched and missing skills, and apply.
- Recruiters can publish jobs, manage applicants, upload résumés in bulk, rank candidates, perform assignments, and form skill-covering teams.
- Advanced Data Structures and Algorithms are used for search, skill extraction, ranking, matching, assigmment, and team formation.

Problem Statement & Objectives

- **Problem:** Manual recruitment and résumé screening are slow, repetitive, inconsistent, and difficult to scale.
- **Objective 1:** Develop an integrated candidate–recruiter talent marketplace.
- **Objective 2:** Automate résumé skill extraction and candidate–job compatibility calculation.
- **Objective 3:** Apply advanced DSA algorithms for searching, ranking, assignment, and team formation.
- **Scope:** Job discovery, résumé ingestion, skill matching, applicant ranking, skill-gap analysis, assignment, and team formation.

Tools & Technologies

- **Front-End:** Thymeleaf, HTML, CSS, Bootstrap
- **Back-End:** Java, Spring Boot, Spring MVC
- **Security:** Spring Security, BCrypt
- **Persistence:** Spring Data JPA, H2 Database
- **Build & Testing:** Apache Maven, JUnit 5, Spring Boot Test, MockMvc
- **Résumé/File Processing:** Apache PDFBox, Apache POI

DSA Algorithms Used

Algorithm	Use / Purpose
KMP	Exact job keyword / marketplace search
Aho–Corasick	Multi-pattern skill extraction from résumé text
Wagner–Fischer Edit Distance	Fuzzy skill matching and typo suggestions
Priority Queue	Candidate/applicant ranking by compatibility score
Hungarian Algorithm	Maximum-score one-to-one candidate–job assignment
Edmonds–Karp	Capacity-aware assignment for jobs with multiple openings
Bitmask Dynamic Programming	Exact minimum-team formation for small skill sets
Greedy Set Cover	Approximate team formation for larger skill sets
Randomized Hashing	Skill-index hashing demonstration and collision analysis
Parallel Processing	Concurrent processing of bulk résumé batches

System Architecture

Candidate Module
- Profile Management
- Browse/Search Jobs
- Compatibility
- Applications

Recruiter Module
- Job Publishing
- Applicant Management
- Bulk Résumé Upload
- Ranking / Selection

Marketplace Module
- Job Listings
- Search
- Matching
- Recommendations

Résumé Processing Module
- PDF / DOCX / TXT / CSV / ZIP
- Text Extraction
- Skill Extraction
- Review / Import

Presentation Layer (Thymeleaf + Bootstrap)

Controller / Application Layer (Spring MVC Controllers + Services)

Algorithm & Index Layer (KMP, Aho–Corasick, Edit Distance, Ranking, Assignment, Team Formation, Skill Indexes)

Persistence Layer (Spring Data JPA + H2 Database)

Workflow / Process Flow

Candidate / Résumé Input → Text & Skill Extraction → Skill Normalization → Compatibility Scoring → Candidate Ranking / Assignment → Recommendations / Recruitment Decision Support

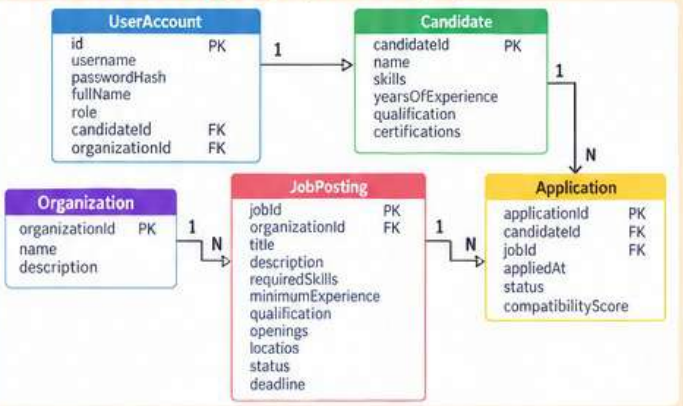
Candidate Flow

Signup / Login → Maintain Profile → Browse Jobs → View Match Score → View Matched & Missing Skills → Apply → Track Application

Recruiter Flow

Signup / Login → Organization → Publish Job → View Applicants → Rank Applicants → Update Application Status → Assignment / Team Formation

Database Design (Simplified ER Diagram)



Core user, candidate and job records are persisted; some marketplace state remains in-memory in the academic prototype.

Key Features

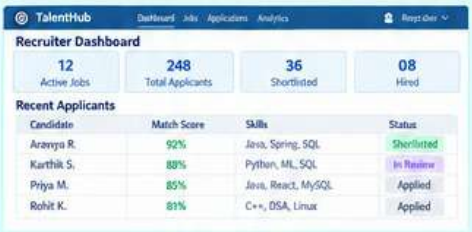
- Candidate and Recruiter signup/login
- Role-based access
- Candidate profile management
- Recruiter job publishing
- Job marketplace and application tracking
- Compatibility score with explanation
- Matched and missing skill analysis
- Bulk résumé import
- PDF, DOCX, TXT, CSV and ZIP support
- Applicant ranking
- Hungarian and Edmonds–Karp assignment
- Bitmask DP and Greedy Set Cover team formation
- Algorithm insight and performance analysis



Implementation Screenshots



Candidate Marketplace Page



Recruiter Dashboard Page



Algorithm Insights / Team Formation Page

Results & Testing

- Automated unit and integration testing implemented
- Candidate–job compatibility and ranking verified
- Duplicate paplication prevention tested
- Résumé parsing and import workflows tested
- Assignment and team-formation algorithms tested
- Maven build and test workflow supported

✓ TESTED WITH MAVEN

Conclusion

- Combines a usable recruitment marketplace with advanced Data Structures and Algorithms.
- Improves efficiency, transparency, and explainability in candidate–job matching.
- Demonstrates practical applications of string algorithms, dynamic programming, graph algorithms, approximation algorithms, hashing, and parallel processing.

Future Scope

- Complete persistence for all marketplace entities
- PostgreSQL / MySQL production database
- OCR support for scanned résumés
- Expanded skill dictionarty and semantic skill recognition
- Configurable compatibility weights
- Email / application-status notifications
- Interview scheduling
- Cloud deployment
- Large-scale analytics and pagination
- Min-cost / weighted flow for capacity-aware score optimization.

References: [1] Cormen, Leiserson, Rivest & Stein – Introduction to Algorithms | [2] Spring Boot Documentation | [3] Spring Security Documentation | [4] Apache PDFBox Documentation | [5] Apache POI Documentation | [6] Java Platform Documentation